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The effect of low-level laser therapy on the success rate of inferior alveolar nerve blocks in mandibular molars with symptomatic irreversible pulpitis: A randomized clinical trial

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INTRODUCTION

It is difficult to achieve complete pulpal anaesthesia in mandibular molar teeth with symptomatic irreversible pulpitis (SIP; Mousavi et al., 2020; Nagendrababu et al., 2020a). It has been reported that the success rate of inferior alveolar nerve blocks (IANB) is between 19% and 56% in these teeth (Farhad et al., 2018). Complete anaesthesia is difficult to achieve for patients with this painful clinical condition for a variety of reasons, such as altered resting potentials of inflamed nerves and decreased excitability thresholds, tetrodotoxin-resistant sodium channels (resistant to anaesthetics), increased expression of sodium channels in teeth with irreversible pulpitis and anxious patients with lower pain thresholds (Wallace et al., 1985). Failure of IANB in mandibular molars with SIP probably involves a more complex neuronal process. Inflammation and bacterial penetration can lead to increased expression of neuropeptides such as substance

P and calcitonin gene-associated peptide, by making nerve fibres more sensitive and increasing the number of impulses. It may also lead to the release of inflammatory mediators such as prostaglandin E2 (PGE2), prostaglandin F2a, interleukins 1 and 6, and tumour necrosis factor. In addition, clinical conditions such as neuronal plasticity, allodynia, peripheral and central hyperalgesia, central sensitization may cause failure of IANB (Hargreaves et al., 2012). These factors may help explain why local anaesthesia is not always effective for mandibular molars with SIP. Many methods have been investigated to IANB success, such as changing the type of anaesthetic solution (Ashraf et al., 2013), the use of different volumes of anaesthetic solution (Fowler & Reader, 2013), buffering the anaesthetic solution (Schellenberg et al., 2015), administering preoperative drugs to patients (Oleson et al., 2010), applying supportive anaesthesia techniques in addition to IANB (Aggarwal et al., 2009), acupuncture (Jalali et al., 2015) and cryotherapy (Topçuoğlu et al., 2019).

Low-level laser therapy (LLLT) has gained attention due to its effectiveness in reducing pain (Mester et al., 1968). Recently, LLLT has been used in Dentistry for a range of treatments, including reduction of orthodontic pain, symptomatic oral lichen planus cases, healing of maxillofacial defects and prophylaxis of stomatitis (Al-Maweri et al., 2017; Oberoi et al., 2014; Ren et al., 2015; Santos et al., 2017). LLLT has begun to be used in endodontic treatments due to its ability to increase wound healing, its role in root canal disinfection, its role in reducing pain and its very limited side effects (Asnaashari et al., 2016; Solmaz et al., 2016). However, the pain reduction mechanism of LLLT is not fully understood. In a systematic review, Chow et al. (2011) suggested that the laser light absorbed by nociceptors may have an inhibitory effect on A & C pain fibres, thus slowing the conduction velocity, reducing the amplitude of the compound effect potential and suppressing neurogenic inflammation. This single-blind randomized controlled clinical trial aimed to evaluate the effect of LLLT on the success of IANB in mandibular molar teeth with SIP.

MATERIALS AND METHODS

The study protocol was approved by the ethics committee (no: 2019/628) and reported using the Preferred Reporting Items for Randomized Trials in Endodontics (PRIRATE) guidelines (Nagendrababu et al., 2020b). A flow chart of the study describes the flow of patients through the trial (Figure 1). The study was registered at www.clinicaltr

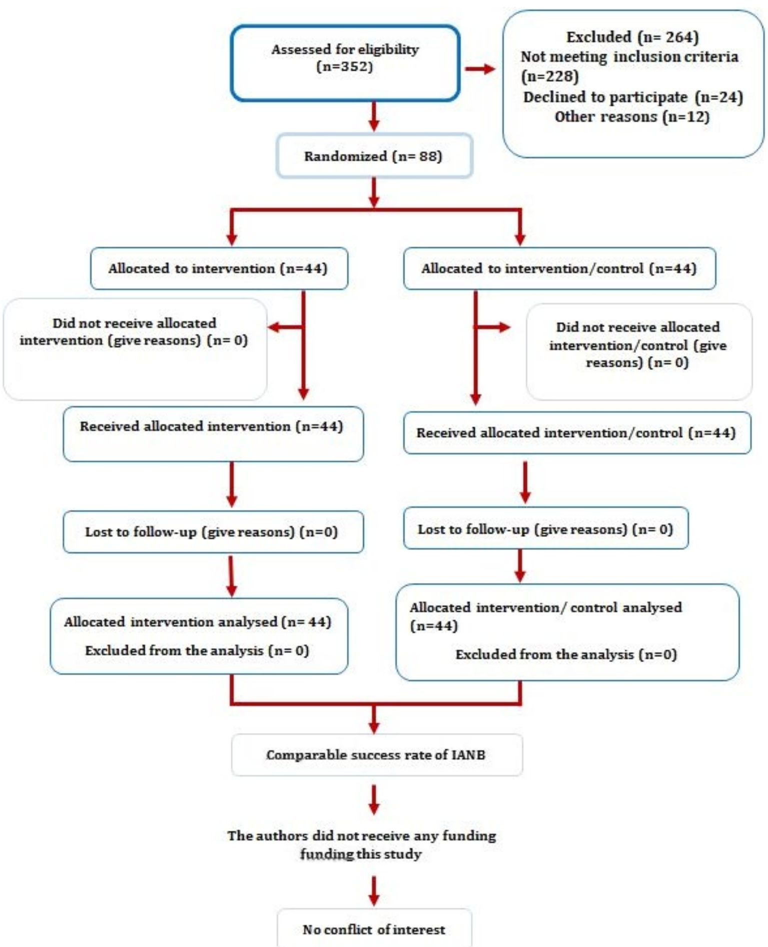

FIGURE 1 Flow chart of participants along the study according to PRIRATE 2020

ials.in.th (Identification number: 20201221002). The number of patients to be included in the study was determined according to a pilot study (with a significance level of $\alpha = 0.05$, power $\beta = 0.80$, and effect size = 0.782). According to the results of the pilot study, it was decided that at least 40 patients should be included in each group. As a result, 88 patients ($n = 44$ for each group) were included to the study (Figure 1).

Inclusion criteria were as follows:

1. Healthy individuals aged between 18 and 65 years
2. Mandibular molar teeth diagnosed with SIP
3. ASA 1–2 patients without systemic disease or with mild systemic disease

The preoperative pain scores of the patients and the pain scores at the time of treatment were recorded on a visual analog scale (VAS). The VAS consisted of a 100-mm horizontal ruler with marks every 10 mm and no numbers except for 0 at the beginning of the scale and 10 at the end of the scale. According to the values recorded on the VAS, the pain levels were classified as no pain (0–4 mm), mild pain (5–44 mm), moderate pain (45–74 mm) or severe pain (75–100 mm; Yaylali et al., 2017). Only patients with moderate or severe pain according to VAS were included. The diagnosis of SIP in teeth included in the study was made using the cold test Green Endo-Ice (1,1,1,2 tetrafluoroethane; Hygenic Corp.) and a digital electrical pulp tester (Parkell Inc.). Pulpal sensitivity was confirmed by a positive response to electric pulp testing and a prolonged response to cold testing. The periapical status was examined using periapical radiographs, and only teeth without periapical radiolucency on the radiographs (except for a widened periodontal ligament with an intact lamina dura) were included.

Exclusion criteria were as follows:

1. Patients who used analgesic or anti-inflammatory drugs within 12 h of treatment
2. Patients who were allergic to nonsteroid anti-inflammatory drugs (NSAI)
3. Pregnant or breastfeeding women
4. Patients taking opioids
5. Individuals with a history of gastrointestinal disease
6. Teeth with periodontal disease
7. Presence of systemic disorders and sensitivity to articaine with 1: 100 000 epinephrine
8. Presence of periapical radiolucency
9. Patients with more than one painful tooth

In addition, teeth without pulpal bleeding after preparing the access cavity were also excluded from the study.

Eighty-eight patients were randomly divided into two groups using a computer random table generator (www.random.org) with a 1:1 allocation ratio by one dental assistant. All operators involved in the study had at least 5 years of experience. The first operator who performed the injections was blinded to the interventions. A second operator performed the LLLT applications. A third operator performed the endodontic treatments; he was not informed of the allocation.

IANB + LLLT group

In this group, a standard IANB was administered to the patients. Patients were slowly injected with 3.6 mL 4% articaine HCI with 1 : 100 000 epinephrine over a period of 1 min. The root canal treatment was then commenced 15 min after the IANB if the patient had lip numbness and two consecutive negative responses to the electric pulp test. Otherwise, the patient was excluded from the study. LLLT was applied to patients who had numbness of the lips. LLLT was performed using a 940 nm wavelength diode laser (Epic 10, Biolase Inc.). The application tip of the laser was placed at a distance of approximately 10 mm from the surrounding tissues of the apices of the mesial and distal root of the teeth (Figure 2). The laser was set to 'Pain therapy' mode and activated at 0.5-watt power. The laser was applied for 30 s in the area of the mesial root of the mandibular molar teeth and 30 s with circumferential movements in the area of the distal root. Then, the access cavity was prepared under rubber dam isolation. The pain value reported by the patient at the time of operation was re-determined using the VAS. Patients who felt pain during access cavity preparation and shaping of the canals were told to mark the pain value they felt on the VAS again. Anaesthesia was categorized as unsuccessful if the pain values felt by the patients at any of these stages were higher than 44 mm in VAS.

IANB group

As in the IANB + LLLT group, a standard IANB was administered to the patients. The root canal treatment was commenced 15 min after the IANB if the patient had lip numbness and two consecutive negative responses to the electric pulp test. Otherwise, the patient was excluded from the study. The laser tip was placed at a distance of approximately 10 mm from the tissue around the apex of the root; however, it was not activated (mock laser application-placebo; Arslan et al., 2017). The access cavity was prepared under rubber dam isolation. The pain value of the patient at the time of operation was re-determined using the VAS. Patients who felt pain during preparation the access cavity and shaping the canals were told to mark

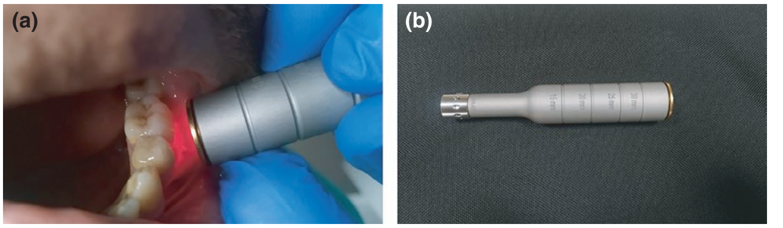

FIGURE 2 Low-level laser application (a); laser tip (b)

TABLE 1 Pain scores after anaesthesia for the groups

	No pain	Mild	Moderate	Severe
Groups				
Inferior alveolar nerve block	7	8	23	6
Inferior alveolar nerve block + low-level laser therapy	12	13	16	3

the pain value they felt on the VAS again. Anaesthesia was categorized as unsuccessful if the pain values felt by the patients at any of these stages were higher than 44 mm in VAS.

In both groups, buccal anaesthesia, intraligamentary anaesthesia or intrapulpal anaesthesia were administered as supportive anaesthesia for patients whose anaesthesia failed and their treatment continued.

Table 1 shows the distribution of the pain scores after anaesthesia for the groups. In the IANB group, the anaesthesia success rate was 34%. For the patients in the IANB + LLLT group, the anaesthesia success rate was 57%. A significant difference was determined between the groups ($p = .032$).

DISCUSSION

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